Siamese network change-detection method

Part 2 · supervised deep learning, trained on image patches

config.yaml (dataset section)

5 regions · bands B2,B3,B4,B8 · patch 256 px @ 10 m · split 70/15/15 by spatial blocks (seed 42).

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Patch export — build_dataset.py / dataset_build.py

256x256 @10 m: before + after reflectance + Hansen label + valid mask -> data/patches/*.npz (2,154 patches).

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Manifest + region means — build_manifest.py

assign train/val/test by whole spatial blocks; write manifest.csv and region_means.json.

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PatchDataset — src/dataset.py

load .npz; scale (raw /10000, OR / region-mean if normalised); augment on train -> before/after/label/valid tensors.

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Siamese U-Net — src/model.py (~7.8 M params)

one shared encoder over before & after -> |feat_before - feat_after| per scale -> U-Net decoder -> per-pixel logits.

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Train — train.py (Google Colab T4 GPU)

masked BCE over valid pixels · pos_weight 15 · Adam 1e-3 · 30 epochs · keep best val-F1 -> runs/siamese_base.pt.

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Evaluate on held-out TEST patches

TP / FP / FN over valid pixels -> P / R / F1 (the SAME metric as the classical method).

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RESULT

Precision / Recall / F1 on held-out patches. Test F1 0.591 baseline vs 0.531 normalised (normalisation does NOT help the deep model).